

ECEN 4273/5080

Project 2

October 10, 2025

In a self organized group of 2 to 4 people, create an object recognition program using a deep learning approach. You may use any coding language and artificial neural network libraries of your choosing with the exception of MATLAB. You may use MATLAB as a learning tool and even model your code off of MATLAB code but you may not use it as your final implementation. The reason for this is, MATLAB is not a generalized solution and depends on the MATLAB engine to run which is neither free nor small. The requirements for your program are the following:

- Your program must be able to recognize that the following objects are in an image to within 75% accuracy. You may assume that only one of these objects are in the image at a time.
 - A Dalek

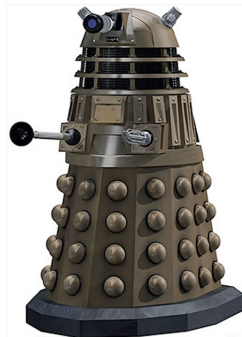


Figure 1: Dalek

- A lightsaber



Figure 2: Lightsaber

- A cat

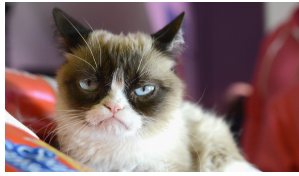


Figure 3: Cat

- A dog

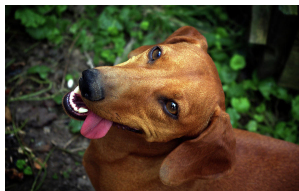


Figure 4: Dog

- A person



Figure 5: Person

- Your program must take as input a short video file (.mp4) and outputs another video file (.mp4) with each of the frames annotated with a bounding box around recognized objects. This operation does not have to be real-time.

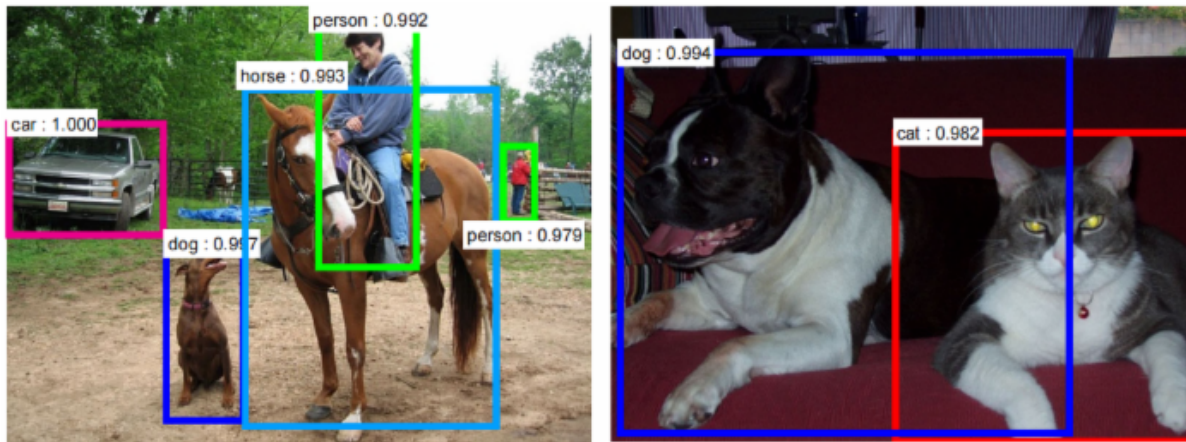


Figure 6: Example of Bounding Box with Labels

You must also implement one of the following options to receive an A:

- Real-time operation with a video stream. In this case, the input and output will not be files. The input will be a video stream and the output will be displayed to the screen with all bounding boxes shown in real-time.
- Capability of identifying multiple objects in the frame at once.
- Accuracy on your test set in excess of 85% for all classifications
- Differentiate between a Sith lightsaber and any other type to within 75% accuracy.



Figure 7: Sith Lightsaber

I suggest you start with an open source object recognition project that uses a library like pytorch, keras, tensorflow, or caffe in conjunction with an image processing library like opencv, Pillow/PIL, or scipy. These tools are much more plentiful when using python. If you are not familiar with python, it is pretty easy to pick up and I can help get you over any hurdles you may have.

Inherent in this project is data processing. You may need to implement an image scraper to build a dataset for images that are not commonly used in object recognition datasets (lightsabers, daleks). You will likely need to process your training images so that they are all of uniform size and resolution to input to your neural network.

Deliverables (at the end of the project):

- Your program, trained and capable of being used immediately after downloading from github.
- A README Document that describes library dependencies, installation and usage procedures.
- Documentation automatically generated from a tool such as pydoc. This documentation can be hosted on your github page so that you need not upload them to canvas.
- A test plan document describing your unit tests and integration tests for the project.
- Test scripts written for a CI/CD tool like Jenkins, Kubernetes, or Travis CI.
- A graph depicting your cost function output as the network is trained for both the training data set and the verification dataset.
- Results from your test dataset showing that each object was recognized to within 75% accuracy.

You are required to use iceScrum to track your project and sprints and you are required to use github to manage your code and documents. One person should act as Scrum Master and one person should be the owner of the Git repo.

A 5 point bonus will be awarded to highest accuracy of object recognition.

Dates:

- Start of First Sprint - Monday Oct. 13
- End of First Sprint - Sunday Oct. 26
- Demo 1 - Monday Oct. 27
- Start of Second Sprint - Monday Oct. 27
- End Of second Sprint - Sunday Nov. 9
- Start of Third Sprint - Sunday Nov. 23
- Final Demo - Wednesday Dec. 3 (possibly Friday Dec. 5th)
- Deliverables Due - Friday Dec. 5th